

CE222	Computer Organization & Assembly Language (3 CH)	Knowledge Profile: WK5	Focus: PBL	CS/CE/AI/DS/ CyS/SE
Pre-Requisite: CE222 Instructor: Dr. Rashid Jillani Office # G-05, FCSE, GIK Institute, Ext. 2520 Email: rjillani@giki.edu.pk Office Hours: 09:30 a.m. ~ 03:00 p.m.				

Course Introduction
The main objective of the course is to introduce the architectural concepts, organization, basic principles and operations of various building blocks of a typical computer system. Emphasis is given to structures, functions and characteristics of computer systems and design of the various functional units and components of computers. Emphasis is also given to expose the low-level logic employed for problem solving while using assembly language as a tool. During the course, a design project based on complex engineering problem is assigned to students for a minimum duration of six weeks.

Course Contents
Broadly, the course covers structural and functional organizations, register transfer and microoperations, basic computer organization and design, assembly language of three-bit opcode processor, programming the basic computer, microprogrammed control, central processing unit, input-output organization, and memory organization.

Mapping of CLOs and PLOs				
Sr. No.	Course Learning Outcomes (CLOs) ⁺	WA PLOs	SA PLOs	Bloom's Taxonomy level (Cognitive domain)
CLO 1	Understand the fundamental knowledge of the basic hardware building blocks from which computers are constructed.	PLO 1	Knowledge for Solving Computing Problems	C2 (Comprehension)
CLO 2	Apply their knowledge of the structure and functions of computer systems and the way they are designed.	PLO 1	Knowledge for Solving Computing Problems	C3 (Application)
CLO 3	Apply their knowledge of the different CPU organizations, addressing modes, and instruction formats to design computer systems.	PLO 3	Design/ Development of Solutions	C3 (Application)
CLO 4	Apply their knowledge of assembly language to create moderately complex assembly language programs.	PLO 3	Design/ Development of Solutions	C3 (Application)
*Please add the prefix "Upon successful completion of this course, students will be able to"				
CLO Assessment Mechanism				

Assessment tools	CLO_1	CLO_2	CLO_3	CLO_4
Assignments	50%	–	30%	30%
Quizzes	50%	30%	40%	–
Midterm Exam	–	40%	–	30%
Final Exam	–	30%	30%	40%

Overall Grading Policy	
Assessment Items	Percentage
Assignments	5%
Quizzes	25%
Midterm Exam	30%
Final Exam	40%

Text and Reference Books
Text books: W. Stallings, "Computer Organization and Architecture: Designing for Performance", 11th Ed., Pearson, 2019. M. Mano, "Computer System Architecture", 3rd Ed., Pearson, 2007. Reference books: D. Patterson and J. Hennessy, "Computer Organization and Design – MIPS Edition: The Hardware/Software Interface", 6th Edition, Morgan Kaufmann, 2021. J. Hennessy & D. Patterson, "Computer Architecture: A Quantitative Approach", 6th Edition, Morgan Kaufmann, 2019.

Administrative Instruction	
▪ According to the Institute’s policy, minimum attendance must be 80% for appearing in final examination. ▪ In any case, there will be no retake of (scheduled/surprise) quizzes. ▪ For queries, kindly follow the office hours in order to avoid any inconvenience.	
Computer Usage/Software Tool	
• Three-bit Opcode Processor simulator and Laurens Rodriguez’s Basic Computer simulator are to be used for the assembly language programming of Three-bit Processor and Basic Computer, respectively.	
Lecture Breakdown	
Week 1	• Architecture and organization, • Digital computer systems, • Structural and functional organizations, • Von-Neumann architecture, • Microoperations, • Register transfer language, • Control functions, • Register transfer microoperations.
Week 2	• Registers transfer, • Microprocessor bus structure, • Bus and memory transfers, • Arithmetic microoperations, • Binary incrementer.
Week 3	• Binary adder, • Binary adder–subtractor, • Arithmetic circuit, • Logic microoperations, • Applications of logic microoperations.
Week 4	• Shift microoperations, • Logical, circular and arithmetic shifts, • Arithmetic logic and shift unit, • Three-bit Opcode Processor.
Week 5	• Assembly language programming of Three-bit Opcode Processor, • Basic Computer memory, • Instruction format, • Addressing modes, • Basic Computer instruction, • Processor registers.
Week 6	• Instruction set completeness, • Common bus system, • Basic computer instruction cycle, • Timing and control, • Control unit of Basic Computer.
Week 7	• Memory reference instructions, • Timing signals, • Register reference instructions, • I/O reference instructions.
Week 8	• Program interrupts, • Interrupt cycle, • Input/out configuration, • Complete computer description, • Control of registers, flags, memory and common bus.
Week 9	• Machine language, • Objectives & perspectives of assembly language, • Pseudo-instructions, • Assembly language programs, • Translation to binary, • Program loops.

Week 10	• Programming arithmetic operations, • Double precision addition, • Programming logic and shift operations, • Programming subroutines, • Input/output programming.
Week 11	• Interrupt programming, • Microprogrammed control organization, • Control memory, • Microprogram and microinstructions, • Instruction formats.
Week 12	• Mapping of instructions, • Symbolic and binary microprograms, • Design of microprogram control unit, • General register organization.
Week 13	• Stack organization, • Reverse Polish notation, • Instruction formats, • Assembly language programs, • Addressing modes, • Data transfer and manipulation, • Program control.
Week 14	• Complex instruction set computers, • Reduced instruction set computers, • Peripheral devices, • Input/output interface, • Isolated and memory mapped I/O, • Serial and parallel priority interrupts.
Week 15	• Modes of transfer, • Direct memory access, • Memory hierarchy, main memory, • Memory address map, • Connection of memory to CPU, • Auxiliary memory.